UNDER REVIEW

RESCIENCE C

Replication / Medical Imaging

[Re] An Open-Source Reproduction and Enhancement of CheXNet for Chest X-ray Disease Classification

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Abstract

Deep learning for radiologic image analysis is a rapidly growing field in biomedical research and is likely to become a standard practice in modern medicine. We used the publicly available NIH ChestX-ray14 dataset, containing 112,120 frontal-view chest X-rays labeled for 14 thoracic pathologies in order to reproduce the 14-pathology extension of CheXNet and built a modified model, DACNet. The reproduced CheXNet model achieved a mean test AUROC of 0.8125, compared with 0.8414 calculated from the 14 per-pathology AUROCs reported in the original article. The modified model, DACNet achieved a mean test AUROC of 0.8474. As a secondary analysis, per-pathology decision thresholds selected only on the validation set produced test macro F1 scores of 0.2813 for the CheXNet reproduction and 0.3414 for DACNet. The original pneumonia-versus-radiologist F1 experiment could not be reproduced because its expert annotations for the 420-image test set are not publicly available.

1 Introduction

Chest X-ray classification is an important task in current medical image analysis. Recently, deep learning models have been trained to detect various thoracic diseases from radiographic scans. CheXNet, a landmark study in this field, introduced a 121-layer DenseNet convolutional neural network that reportedly outperformed radiologists specifically in pneumonia detection[1]. The researchers used the NIH ChestX-ray14 dataset, which is publicly available and has over 100,000 frontal-view chest X-rays labeled with up to 14 disease classes[2]. The success of CheXNet has inspired further research because it demonstrated a significant step toward using artificial intelligence to assist in clinical diagnosis, especially in regions where access to licensed radiologists is limited[3]. In the midst of a reproducibility crisis in academia, independent researchers must reproduce groundbreaking studies like this in order to protect the accuracy of future research[4].

Researchers in this project set out to replicate the original CheXNet model as closely as possible and evaluate AUC-ROC scores across all 14 disease classes. All of the code for our models and our evaluation pipeline is publicly available in our GitHub repository. While our primary goal was to reproduce the original CheXNet study, we also recognize the importance of its successor, CheXNeXt, which validated a similar model against board-certified radiologists on a curated internal dataset[5]. Although that expert-labeled test set is not currently publicly available, its findings emphasize the

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The authors have declared that no competing interests exist.
Code is available at https://github.com/dstrick17/DacNet..
Data is available at https://www.kaggle.com/datasets/nih-chest-xrays/data.
Open peer review is available at https://github.com/ReScience/submissions/issues/98.

clinical relevance of these models and continue to reinforce the need for reproducible benchmarking on public datasets such as NIH ChestX-ray14.
Our first contribution is our reproduction of the 14-pathology CheXNet experiment using ImageNet-pretrained DenseNet-121, a patient-wise split, and the training details reported in the original article. Our second contribution is DACNet, an extension that retains DenseNet-121 while incorporating focal loss, the AdamW optimizer, color jitter, and validation-selected per-pathology F1 thresholds. Unlike the original CheXNet study, which reported an F1 score for pneumonia using an expert-labeled subset, our secondary analysis computes per-pathology F1 scores across all 14 diseases using only the public NIH labels.

2 Related Work

Deep learning methods have proven to achieve substantial accuracy in image classification. In 2012, AlexNet demonstrated the effectiveness of convolutional neural networks on large-scale image classification tasks[6]. Our main source of inspiration is "CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning." In addition to CheXNet, the authors later introduced CheXNeXt, a related model evaluated directly against radiologists on 14 thoracic diseases. Their results showed radiologist-level performance on most conditions. However, the curated validation set used in CheXNeXt is not publicly available, making full replication difficult.
Despite substantial advancements in medical imaging, there is still room for improvement[7]. Other studies have reported the efficacy of deep learning in chest X-ray classification[8], ophthalmology [9], and fracture detection in musculoskeletal radiographs [10].

3 Scope of the Reproduction

The original CheXNet article reports two related but also distinct experiments. The primary experiment is a binary pneumonia detection. This primary experiment uses a single-output DenseNet-121, a class-weighted binary cross-entropy loss, and an expert-labeled test set of 420 images to compare the model with four radiologists. The second experiment extends the network to all 14 pathologies by replacing the final layer with 14 outputs and optimizing the sum of unweighted binary cross-entropy losses. It uses a 70%/10%/20% patient-wise split and reports per-pathology AUROC.
This study reproduces the second 14-pathology experiment. The distinction is important: the weighted loss and reported pneumonia F1 comparison belong to the first binary pneumonia experiment. We cannot reproduce that comparison because the expert annotations for its 420-image test set were not released. We therefore treat the 14 per-pathology AUROCs as the primary reproduction targets. The F1 scores reported here are a secondary analysis using public NIH labels and are not presented as a reproduction of the radiologist comparison.
We first contacted the email address listed in the original paper, which was associated with Stanford University. After receiving no response, we learned that the author had moved to Harvard Medical School and contacted his current laboratory in October 2026 to request access to the expert annotations. We have not yet received a response.

4 Methods

4.1 Dataset and patient split

All data used in this study came from the NIH ChestX-ray14 dataset, the same dataset used to build the CheXNet algorithm. The dataset is available through the Kaggle NIH Chest X-rays page[2] and the NIH distribution [11]. It contains 112,120 frontal-view chest X-rays from 30,805 patients. Each image is labeled with "No Finding" or with one or more of 14 pathologies. The labels were extracted automatically from radiology reports and therefore contain label noise.

One major challenge is the extreme class imbalance within the dataset. While there are only 14 possible classifications, any given image can exhibit any combination of these diseases. This creates a theoretical space of 2^{14}, or 16,384 possible label combinations. Exploratory data analysis found 836 observed combinations. Approximately 54% of the images are labeled "No Finding," and many individual combinations occur rarely. This imbalance presents a substantial challenge for threshold-dependent metrics such as F1.

We sorted the unique patient identifiers and created a deterministic split with seed 42. The training, validation, and held-out test sets contain 70%, 10%, and 20% of patients, respectively, with no patient overlap. This produced 21,563 training patients (78,614 images), 3,081 validation patients (11,212 images), and 6,161 test patients (22,294 images). The exact assignments are stored in `patient_splits.csv` under `reproducibility/replicate_chexnet` and were shared by both models.

The original article states the same 70%/10%/20% patient-level proportions for the 14-pathology experiment but does not publish the exact patient assignments. Our seeded split therefore reproduces the stated procedure rather than the authors' exact partition.

4.2 CheXNet reproduction

We initialized a DenseNet-121 with ImageNet weights and replaced its classifier with a fully connected layer producing 14 logits. Images were converted to RGB, resized to 224×224, converted to tensors, and normalized using the ImageNet channel means and standard deviations. Random horizontal flipping was applied during training. These choices follow the preprocessing described in the original CheXNet article.

Following the original 14-pathology extension, the loss was the sum of 14 unweighted binary cross-entropy losses, averaged over the minibatch. We used Adam with $\beta_1 = 0.9$, $\beta_2 = 0.999$, an initial learning rate of 0.001, and a batch size of 16. The learning rate was reduced by a factor of 10 whenever validation loss failed to improve after an epoch. Training was configured for at most 20 epochs with early stopping after five epochs without improvement, and the checkpoint with the lowest validation loss was selected.

4.3 DACNet extension

DACNet retains the ImageNet-pretrained DenseNet-121 backbone and 14-output classifier. It replaces binary cross-entropy with focal loss ($\gamma = 2$, $\alpha = 1$) to address class imbalance and uses AdamW with a learning rate of 0.00005 and weight decay of 10^{-5}. A `ReduceLROnPlateau` scheduler reduces the learning rate by a factor of 10 after one epoch without improvement in validation loss. The model was trained for at most 25 epochs with a batch size of 8 and early stopping after five epochs without improvement in mean validation AUROC. The checkpoint with the highest mean validation AUROC was selected.

DACNet training augmentation consisted of a random resized crop to 224×224, random horizontal flipping, and color jitter with brightness and contrast factors of 0.1. Validation and test images were resized to 256 pixels and center-cropped to 224×224. All

images used ImageNet normalization. DACNet used the exact patient assignments generated for the CheXNet reproduction.

4.4 Evaluation

The primary outcome for the reproduction study was the area under the receiver operating characteristic curve (AUROC), which was calculated independently for each pathology on the held-out test set. Mean AUROC was calculated as the unweighted mean of the 14 pathology-level values. We compared these results with the values reported in the original CheXNet article.
The F1 scores require a decision threshold and are reported as a secondary analysis. For each pathology, we selected the threshold that maximized F1 on the validation set and then applied that threshold unchanged to the held-out test set. We report the unweighted macro average across all 14 pathology-level F1 scores. For transparency, the CheXNet reproduction also records its test F1 at a fixed threshold of 0.5. No threshold was selected using the test labels.
The two models use differently scaled loss functions, so their numerical test losses are not directly comparable. We therefore compare the models using AUROC and F1 rather than raw loss values.

5 Results

5.1 Reproduction of the 14-pathology AUROC results

The reproduced CheXNet model achieved a mean test AUROC of 0.8125. The mean of the 14 per-pathology values reported in the original article is 0.8414. The reproduction exceeded the corresponding published AUROC for two pathologies, Consolidation and Effusion, and was lower for the other 12 pathologies. Table 1 reports the complete comparison.
Our reproduction fell short of the published mean AUROC by 0.029 and was lower for 12 of the 14 pathologies. As a result, we consider it as a partial reproduction rather than an exact reproduction. The overall pattern was recovered, but the numerical values were not. We cannot identify the cause of the gap from a single run. However, there are several plausible explanations that follow from differences between our setup and the original. First, the original patient-level assignment was not published, so our seeded 70%/10%/20% split necessarily places different patients in the test set; because many pathologies are rare, small changes in which positive cases fall in the test set can shift per-pathology AUROC noticeably. Additionally, neural network training is stochastic, and we completed only one run per model, so some of the difference may be ordinary seed-to-seed variation that we did not measure. Third, our software libraries and hardware differ from those available in 2017, which can alter pretrained-weight handling, numerical behavior, and data-loading details. Finally, the NIH labels are extracted automatically from reports and are noisy, which may make results more sensitive to the particular split. The gap was largest for Hernia, Emphysema, Mass, Nodule, and Pneumothorax, but we did not test whether any of these explanations accounts for it.

5.2 DACNet extension

DACNet achieved a mean test AUROC of 0.8474 and exceeded the published CheXNet AUROC for 8 of the 14 pathologies. The largest numerical gains over our reproduction occurred for Hernia, Fibrosis, Nodule, Emphysema, and Mass. Because this study contains one completed training run for each model and no confidence intervals or repeated-seed analysis, these differences are descriptive and should not be interpreted as evidence of statistical significance.

Pathology	Original CheXNet	Reproduction	DACNet
Atelectasis	0.809	0.790	**0.828**
Cardiomegaly	**0.925**	0.917	0.913
Consolidation	0.790	0.800	**0.824**
Edema	0.888	0.883	**0.895**
Effusion	0.864	0.874	**0.883**
Emphysema	**0.937**	0.877	0.920
Fibrosis	0.805	0.782	**0.827**
Hernia	0.916	0.847	**0.951**
Infiltration	**0.735**	0.695	0.712
Mass	**0.868**	0.816	0.862
Nodule	0.780	0.722	**0.794**
Pleural thickening	0.806	0.781	**0.808**
Pneumonia	**0.768**	0.739	0.763
Pneumothorax	**0.889**	0.850	0.884
Mean	0.841	0.812	**0.847**

Table 1. Held-out test AUROC by pathology. Original values are those reported by Rajpurkar et al. The bold value in each row is the largest of the three.

Since our CheXNet reproduction fell short of the published results, we wanted to build a model that improved on it and so future researchers could continue to iterate on. We chose modifications aimed at the class imbalance described above. Most images in the NIH dataset are labeled ”no finding” and several pathologies are rare, so we expected the unweighted binary cross-entropy loss to be dominated by easy negative examples. Focal loss down-weights well-classified examples and was a natural candidate for this setting. We chose AdamW because its decoupled weight decay offers a simple form of regularization on a network with a large number of parameters, and color jitter was added as a mild augmentation to reduce overfitting to the intensity variations between scans. These choices were exploratory and were not tuned individually. By documenting them openly, we hope to give future open-source researchers a concrete starting point for further improvements.

5.3 Secondary F1 analysis

At the fixed threshold of 0.5, the CheXNet reproduction had a test macro F1 of 0.1271. Selecting one threshold per pathology on the validation set increased its held-out test macro F1 to 0.2813. Using the same validation-only threshold selection procedure, DACNet achieved a test macro F1 of 0.3414. The pathology-level values are shown in Table 2. These F1 values do not reproduce the F1 score reported for the original binary pneumonia model. The original value was calculated on expert annotations for a different model and a 420-image test set that is not publicly available. The present values instead describe thresholded performance of the 14-output models against the public NIH labels.

6 Reproducibility and Limitations

The repository provides the training scripts, dependency files, exact patient assignments, machine-readable result files, and commands for local and Kaggle execution. The NIH dataset can be attached directly to a Kaggle notebook, so a reviewer does not need to download all images to a personal computer. A short smoke test checks dataset discovery, model initialization, training, evaluation, and result output; it is not intended to reproduce the final metrics. Full runs require GPU computation.

The exact patient assignment and trained weights from the original CheXNet article were not available. Although our split follows the reported 70%/10%/20% patient-level

Pathology	CheXNet reproduction	DACNet
Atelectasis	0.365	**0.413**
Cardiomegaly	**0.401**	0.394
Consolidation	0.228	**0.245**
Edema	0.231	**0.240**
Effusion	0.516	**0.531**
Emphysema	0.363	**0.472**
Fibrosis	0.131	**0.165**
Hernia	0.103	**0.444**
Infiltration	0.404	**0.424**
Mass	0.320	**0.397**
Nodule	0.233	**0.321**
Pleural thickening	0.200	**0.225**
Pneumonia	0.077	**0.086**
Pneumothorax	0.367	**0.422**
Macro average	0.281	**0.341**

Table 2. Held-out test F1 using pathology-specific thresholds selected only on the validation set.

procedure, it is not necessarily identical to the original split. The public NIH labels were extracted automatically from reports and may contain errors. The expert annotations required for the original pneumonia-versus-radiologist comparison are also unavailable. Finally, our comparison is based on a single training run per model, so it does not quantify variation across random seeds.

7 Discussion

Simply reproducing CheXNet's 14-output structure did not yield the exact published AUROC values. Nevertheless, the reproduction recovered the overall pattern of performance: Cardiomegaly, Effusion, Emphysema, and Pneumothorax were among the stronger classes, while Infiltration, Nodule, and Pneumonia were more difficult. The lower mean AUROC demonstrates why implementation details, data partitions, and saved artifacts are important when evaluating a replication.

DACNet produced a higher mean AUROC and validation-thresholded macro F1 than our CheXNet reproduction on the same patient partition. However, the study does not isolate the effect of each modification. DACNet should therefore be interpreted as an exploratory extension rather than as part of the CheXNet reproduction itself.

We are unsure exactly why our AUROC values differ from the original CheXNet results. It is possible that future reproduction attempts may achieve results closer to those originally reported, or they may produce results more consistent with our reproduction. The F1 scores cannot be reliably compared since the original study used an expert-annotated subset for the pneumonia-versus-radiologist comparison, which is not available to us. However, the F1 scores we calculated were still significantly lower and not at expert radiologist levels.

8 Conclusion

This project aimed to reproduce and extend CheXNet, a landmark deep learning model in medical imaging. We reproduced the public 14-pathology experiment and obtained a mean held-out test AUROC of 0.8125, below the 0.8414 mean calculated from the original article's reported values. We could not reproduce the separate pneumonia-versus-radiologist F1 experiment because its expert labels are not public. DACNet, an exploratory extension using focal loss, AdamW, color jitter, and validation-selected

thresholds, achieved a mean test AUROC of 0.8474 and a macro F1 of 0.3414 on the same patient split.
In addition to the technical contributions, this project emphasizes the importance of reproducibility in machine learning research. Our primary motivation was not only to evaluate the claims of a high-impact study, but also to contribute openly to the scientific community. Our codebase, exact patient split, evaluation tools, and machine-readable results have been made publicly available in hopes of promoting transparency and enabling further development by others.

9 Funding and Competing Interests

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10 Generative AI Usage Statement

Some language in this manuscript was improved using generative AI tools to clarify sentence structure and enhance readability. All AI-generated suggestions were critically reviewed and edited by the authors to ensure accuracy and appropriateness.

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